PROBLEM 10 | Problems T2_CrystStruct.pdf

Crystalline Structures & Miller Indices

Identification of Gallium in an Orthorhombic Cell and Close-Packed Directions

Original statement · Problems T2_CrystStruct.pdf, p. 2

10. Three distinct crystallographic planes in the unit cell of a hypothetic metal are shown below. The circles represent atoms. ***Plane*** (110)*: Rectangle with dimensions* 4.88 Å *(diagonal) and* 3.32 Å *(vertical edge). Atoms at the 4 corners and 2 in the center of the base edges.*
Plane (101)*: Rectangle with dimensions* 4.64 Å *and* 3.65 Å*. Atoms only at the 4 corners.*
Plane (020)*: Rectangle with dimensions* 3.24 Å *and* 3.32 Å*. Atoms at the 4 corners.*
a) *To which crystalline structure does this unit cell belong?* ***b)*** *Given that the density of this metal is* $5.9\ \mathrm{g/cm^3}$*, determine the atomic mass. Which specific metal is it referred to?* ***c)*** *Identify the directions and planes of maximum packing. (Solution: a) Base centered orthorhombic lattice; b)* $M = 69.76\ \mathrm{g/mol}$*, Gallium; c)* $[110], [\bar{1}10], [\bar{1}\bar{1}0], [1\bar{1}0], (001)$*)*

Solution: a) Base centered orthorhombic lattice; b) $M = 69.76\ \mathrm{g/mol}$**, Gallium; c)** $[110], [\bar{1}10], [\bar{1}\bar{1}0], [1\bar{1}0], (001)$

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Dimensions read from the planes:
- From the (020) plane: edge $a = 3.24$ Å, edge $c = 3.32$ Å.
- From the (110) plane: vertical edge $c = 3.32$ Å, base diagonal $\sqrt{a^2 + b^2} = 4.88$ Å.
- From the (101) plane: diagonal $\sqrt{a^2 + c^2} = \sqrt{3.24^2 + 3.32^2} = \sqrt{10.50 + 11.02} = \sqrt{21.52} \approx 4.64$ Å, and edge $b = 3.65$ Å.
- Volumetric density: $\rho = 5.9\ \mathrm{g/cm^3}$.
- Avogadro number: $N_A = 6.022 \times 10^{23}\ \mathrm{atoms/mol}$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Lattice Parameters (a, b, c):**
2. $a = 3.24$ Å
3. $b = 3.65$ Å (check: $\sqrt{3.24^2 + 3.65^2} = \sqrt{10.50 + 13.32} = \sqrt{23.82} = 4.881$ Å, in agreement with the 4.88 Å read from the figure).
4. $c = 3.32$ Å
5. Since $a \neq b \neq c$ and the planes form angles of $90°$, the system is **Orthorhombic** [Session 3 Slide 13].
6. **Cell Centring Type (Bravais) [Session 3 Slides 10, 13]:**

The (110) plane contains atoms at the centre of the lower and upper base edges, which reveals that the basal face (001) has a centred atom. By contrast, the (101) and (020) planes contain no atoms centred on their faces. Therefore, only the two (001) bases are centred $\implies$ **Base-Centred Orthorhombic Lattice (Base-Centered Orthorhombic, C).**

7. **Number of Atoms per Cell:**

$$n = 8 \times \left(\frac{1}{8}\right)_{\text{vertices}} + 2 \times \left(\frac{1}{2}\right)_{\text{bases}} = 1 + 1 = 2 \text{ atoms/cell}$$

8. **Molar Atomic Mass:**

$$M = \frac{\rho \cdot (a \cdot b \cdot c) \cdot N_A}{n}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Identification of the Crystal Lattice

- Edges: $a = 3.24$ Å, $b = 3.65$ Å, $c = 3.32$ Å ($a \neq b \neq c$).
- Angles: $\alpha = \beta = \gamma = 90°$.
- Atom arrangement: Atoms at the 8 vertices and at the barycentres of the two opposite basal faces $z = 0$ and $z = c$.
- Structure: **Base-centred orthorhombic lattice (Base-centered Orthorhombic)**.

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2. Part b: Atomic Mass and Identification

- **Unit cell volume:**

$$V_C = a \cdot b \cdot c = (3.24 \times 10^{-8}\ \text{cm}) \times (3.65 \times 10^{-8}\ \text{cm}) \times (3.32 \times 10^{-8}\ \text{cm}) = 3.9262 \times 10^{-23}\ \text{cm}^3$$

- **Atomic mass (M):**

$$M = \frac{\rho \cdot V_C \cdot N_A}{n} = \frac{(5.9\ \text{g/cm}^3) \times (3.9262 \times 10^{-23}\ \text{cm}^3) \times (6.022 \times 10^{23}\ \text{mol}^{-1})}{2}$$

$$M = \frac{2.3165 \times 10^{-22} \times 6.022 \times 10^{23}}{2} = \frac{139.50}{2} = \mathbf{69.75}\ \text{g/mol}$$

Discrepancy with the official solution. The official key gives $69.76\ \text{g/mol}$. With the $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$ used throughout this page the result is $69.75\ \text{g/mol}$; the key's value follows from $N_A = 6.023 \times 10^{23}$ (the value supplied in Problem 5 of the same sheet), 0.015% higher. The difference is a constant-rounding effect and no data were altered. Both identify gallium ($69.72\ \text{g/mol}$).

- **Identification of the Metal:**

The chemical element with a molar mass of $\approx 69.7\ \text{g/mol}$ and density $\approx 5.9\ \text{g/cm}^3$ is **Gallium** (Ga).

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3. Part c: Directions and Planes of Maximum Packing

- **Plane of Maximum Packing:**
- In the base-centred orthorhombic cell, the basal face (001) contains the centred atom and the 4 vertex atoms over an area $a \cdot b = 3.24 \times 3.65 = 11.83\ \text{Å}^2$.
- Planar density of (001): $\rho_{(001)} = \frac{2}{11.83 \times 10^{-16}\ \text{cm}^2} = 1.69 \times 10^{15}\ \text{at/cm}^2$.
- Therefore, the most densely packed plane is (001) (the basal plane).
- **Directions of Maximum Packing:**
- In the (001) plane, the vertex atoms and the central atom are aligned along the diagonals of the basal face:

$$[\mathbf{110}], \quad [\bar{\mathbf{1}}\mathbf{10}], \quad [\bar{\mathbf{1}}\bar{\mathbf{1}}\mathbf{0}], \quad [\mathbf{1}\bar{\mathbf{1}}\mathbf{0}]$$

- Along these 4 coplanar diagonal directions, the atoms are in continuous mutual contact.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Physics of Gallium:** Pure gallium is an extremely singular element in aerospace materials science and optoelectronics (GaAs, GaN). It has an extremely low melting temperature ($T_m = 29.76^\circ\mathrm{C}$), melting with the heat of a human hand, and crystallises at room temperature in a slightly distorted orthorhombic lattice in which the atoms form almost molecular Ga_2 dimers oriented along the dense directions of the (001) plane.